Unit 1 - Math Fundamentals
Sets

MSDS 520

Meharry Medical College

Outline

Why Sets?

A Problem to Motivate Us

A question about a patient population

"How many patients have diabetes **or** *hypertension, but* **not both***?"*

To answer this we need language for:

- a **collection** of patients ("the diabetics"),
- **combining** collections ("diabetic or hypertensive"),
- **removing** one collection from another ("but not both"),
- and **counting** what is left.

That language is **set theory**. It is the same logic as Unit 1-1 — AND, OR, NOT — but now applied to collections of things rather than to statements.

Where This Shows Up in Data Science

Sets are everywhere once you look

- **Cohorts and subgroups:** every "group of patients" is a set.
- **Joining data:** merging two tables is an intersection of their keys.
- **Domains of functions:** the set of inputs a function accepts.
- **Reference ranges:** "normal" lab values are an **interval** — a set of numbers.
- **Probability:** an event is a set of outcomes (Unit 4).

What we will build

1. **Sets**, membership, and **subsets**.
2. The **operations**: union, intersection, complement.
3. **Venn diagrams** and counting.
4. The standard **sets of numbers** and **intervals**.
5. **Ordered pairs** and tuples — where sets meet coordinates.

Sets and Membership

What Is a Set?

Set

A **set** is a collection of objects, called its **elements** (or **members**).

- Sets are usually named with capital letters: A, B, C.
- Elements are listed inside curly braces:

$$A = \{1, 2, 3\}$$

- **Order does not matter**: $\{1, 2, 3\}$ and $\{3, 1, 2\}$ are the same set.
- **Repeats do not matter**: $\{1, 1, 2\}$ is just $\{1, 2\}$.

A set is defined entirely by *what is in it* — nothing else.

Membership

To say that an object is in a set, we use the symbol $\in$:

$$1 \in A \qquad \text{“1 is an element of } A\text{”}$$

And to say it is not, we strike it through:

$$5 \notin A \qquad \text{“5 is not an element of } A\text{”}$$

Connection to Unit 1-1

“$x \in A$” is a **statement** — it is true or false, and so it can be combined with $\neg$, $\wedge$, $\vee$ like any other. Notice that its truth depends on which x you pick — which is exactly what set-builder notation will use. Everything we do with sets will mirror something we did with logic.

Some Special Sets

Singletons

A set with exactly one element, e.g. $\{1\}$, is called a **singleton**.

Note that 1 and $\{1\}$ are different things: a number, and a box containing that number.

The empty set

The set with **no** elements is called the **empty set**:

$$\emptyset = \{\ \}$$

A universe

Often we fix a **universal set** U containing everything under discussion — e.g. all patients in the study. "Everything else" always means "everything else in U."

Set-Builder Notation

Listing elements only works for small sets. For large or infinite ones, we describe the **rule** that members satisfy:

$$A = \{\, x \mid x > 0 \,\} \quad \text{“the set of all positive numbers”}$$

Read the vertical bar (pipe) as **“such that”**:

“the set of all x *such that* x is greater than 0”

More examples

- $\{\, x \mid x \text{ is a day of the week} \,\}$
- $\{\, x \mid x \text{ is a patient and } x \text{ is diabetic} \,\}$
- $\{\, x \in U \mid x \text{ is diabetic} \,\}$ (membership can be stated up front)

The rule after the bar is a **statement about** x — logic again.

Exercise: Translating Set-Builder Notation

Part 1 — translate set-builder notation into plain English

1. $\{\, x \mid x < 0 \,\}$
2. $\{\, x \in U \mid x \text{ is pregnant} \,\}$ $\quad$ ($U =$ all patients in the study)
3. $\{\, x \mid x \text{ is a month with exactly 30 days} \,\}$ — then list it.

Part 2 — now go the other way, English into set-builder notation

1. The set of all patients who are diabetic **and** over 65.
2. The set of all letters that are vowels.
3. The set of all numbers between 10 and 20.

Exercise: Solution

Part 1 — into English

1. "the set of all x *such that* x is less than 0" — the negative numbers.
2. "the set of all patients in the study who are pregnant."
3. "the set of all months with exactly 30 days"
   $= \{\text{April}, \text{June}, \text{September}, \text{November}\}$.

Part 2 — into set-builder

1. $\{\, x \in U \mid x \text{ is diabetic } \wedge \ x \text{ is over } 65 \,\}$ (the AND from Unit 1-1)
2. $\{\, x \mid x \text{ is a vowel} \,\} \ = \ \{a, e, i, o, u\}$
3. $\{\, x \mid 10 < x < 20 \,\}$

That last one is slippery: "between" never says whether 10 and 20 themselves count. **Intervals** will pin this down.

Cardinality

Cardinality

The **cardinality** of a set A, written $|A|$, is the number of elements it contains.

Examples

- $A = \{1, 2, 3\} \Rightarrow |A| = 3$
- $|\{7\}| = 1$
- $|\emptyset| = 0$

Every "how many patients ..." question is a cardinality question. We will come back to this when we count with Venn diagrams.

Exercise

Try it

Let

$$A = \{2, 4, 6, 8\}$$
$$B = \{\, x \mid x \text{ is a letter in the word "level"} \,\}$$

1. Write B by listing its elements.
2. Find $|A|$ and $|B|$.
3. True or false: $4 \in A$, $\{4\} \in A$, $5 \notin A$.

Exercise: Solution

1. Listing B

The letters of "level" are l, e, v, e, l — but **repeats do not matter**, so $B = \{\mathsf{l}, \mathsf{e}, \mathsf{v}\}$.

2. Cardinality

$|A| = 4$ and $|B| = 3$ — three *distinct* letters, not five.

3. True or false

- $4 \in A$ — **true**, 4 is one of the listed elements.
- $\{4\} \in A$ — **false**. The elements of A are numbers; $\{4\}$ is a *set*, and no set appears in A.
- $5 \notin A$ — **true**, 5 is not listed.

Subsets

Subsets

Subset

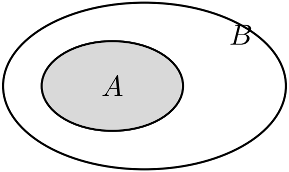
A is a **subset** of B — written $A \subseteq B$ — if **every** element of A is also an element of B.

Written with the implication from Unit 1-1 — for every x:

$$x \in A \ \ \rightarrow \ \ x \in B$$

Example

$\{1, 2\} \subseteq \{1, 2, 3\}$, and $\{1, 2, 3\} \subseteq \{1, 2, 3\}$ — a set is a subset of itself.



Proper Subsets, and the Empty Set

Proper subset

A is a **proper** subset of B — written $A \subset B$ — if $A \subseteq B$ **and** $A \neq B$. That is, B has at least one element that A does not.

$A \subseteq B$	contained in, possibly equal
$A \subset B$	contained in, definitely smaller

Convention: $\emptyset \subseteq B$ for every set B

The empty set is a subset of **everything**.

Why? To fail, $\emptyset$ would need an element that is not in B — and it has no elements at all to offer. The condition is **vacuously true** (Unit 1-1).

Set Equality

Two sets are equal when each contains the other

$$A = B \quad \text{means} \quad A \subseteq B \ \text{ and } \ B \subseteq A$$

This is the set-theory version of the bi-implication $\leftrightarrow$ from Unit 1-1: $x \in A$ if and only if $x \in B$.

This is also a proof strategy

To prove two sets are the same, prove **both** containments:

1. Take an arbitrary $x \in A$ and show $x \in B$.
2. Take an arbitrary $x \in B$ and show $x \in A$.

Exercise

Try it

Let $A = \{1, 2, 3\}$.

1. List **all** subsets of A. How many are there?
2. Which of those are **proper** subsets?
3. True or false: $\emptyset \subseteq A$, $\emptyset \in A$, $A \subseteq A$, $A \subset A$.

Exercise: Solution

1. All subsets — eight of them

$$\emptyset, \quad \{1\}, \{2\}, \{3\}, \quad \{1,2\}, \{1,3\}, \{2,3\}, \quad \{1,2,3\}$$

A set with n elements has 2^n subsets — here $2^3 = 8$.

2. Proper subsets

Every one of them except $\{1,2,3\}$ itself — **seven**.

3. True or false

- $\emptyset \subseteq A$ — **true**. The empty set is a subset of everything.
- $\emptyset \in A$ — **false**. The elements of A are $1, 2, 3$; $\emptyset$ is not one of them.
- $A \subseteq A$ — **true**. $A \subset A$ — **false**: proper demands $A \neq A$.

The Power Set

The eight subsets we just listed are objects in their own right — so we can collect them into a set.

Power set

The **power set** of A, written $\mathcal{P}(A)$, is the set of **all** subsets of A:

$$\mathcal{P}(A) = \{\, S \mid S \subseteq A \,\}$$

Its elements are themselves **sets**, not the original elements.

Example: $A = \{1, 2, 3\}$

$$\mathcal{P}(A) = \big\{\, \emptyset,\ \{1\}, \{2\}, \{3\},\ \{1,2\}, \{1,3\}, \{2,3\},\ \{1,2,3\} \,\big\}$$

$|\mathcal{P}(A)| = 8$, and both $\emptyset$ and A itself are members.

Why 2^n? One yes/no choice per element — in or out — gives 2^n subsets — the same count as the rows of an n-variable truth table (Unit 1-1). **Watch the symbols:** $\{1\} \subseteq A$, but $\{1\} \in \mathcal{P}(A)$.

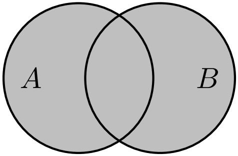
Set Operations

Union

Union

The **union** of A and B, written $A \cup B$, is the set of everything in A, in B, or in both:

$$A \cup B = \{ \, x \mid x \in A \ \textbf{ or } \ x \in B \, \}$$



$\{1, 2, 3\} \cup \{2, 3, 4\} = \{1, 2, 3, 4\}$

Note

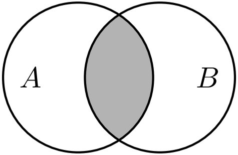
Union is the set version of the logical **OR** — and it is the *inclusive* OR: elements in both sets are included, listed once.

Intersection

Intersection

The **intersection** of A and B, written $A \cap B$, is the set of everything in **both**:

$$A \cap B = \{\, x \mid x \in A \ \ \textbf{and} \ \ x \in B \,\}$$



$\{1, 2, 3\} \cap \{2, 3, 4\} = \{2, 3\}$

Note

Intersection is the set version of the logical **AND**.

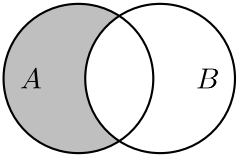
If $A \cap B = \emptyset$, the sets share nothing and are called **disjoint**.

Difference (Relative Complement)

Set difference

The **difference** $A \setminus B$ is everything in A that is **not** in B:

$$A \setminus B = \{\, x \in A \mid x \notin B \,\}$$



Note

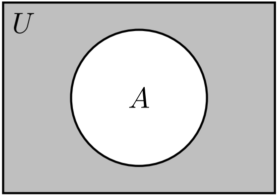
Difference is the set version of **AND NOT**. **Order matters**:

$$\{1, 2, 3\} \setminus \{2, 3, 4\} = \{1\}$$
$$\{2, 3, 4\} \setminus \{1, 2, 3\} = \{4\}$$

Complement

When a universal set U is fixed, the **complement** of A is everything in U outside of A:

$$\bar{A} = U \setminus A = \{ x \in U \mid x \notin A \}$$



Some texts write A' instead of $\bar{A}$.

Note

Complement is the set version of **NOT**.

It only makes sense once U is agreed upon — "not diabetic" means nothing until we say "among whom."

Worked Examples

Let $A = \{1, 2, 3\}$ and $B = \{2, 3, 4\}$.

- $A \cup B = \{1, 2, 3, 4\}$
- $A \cap B = \{2, 3\}$
- $A \setminus B = \{1\}$
- $B \setminus A = \{4\}$

Exercise

Try it

With $A = \{1, 2, 3\}, \quad B = \{2, 3, 4\}, \quad$ and $C = \{3, 4, 5\}$, find:

$$A \cap C, \qquad (A \cup B) \cap C, \qquad A \setminus (B \cap C)$$

Exercise: Solution

$A = \{1, 2, 3\}, \quad B = \{2, 3, 4\}, \quad C = \{3, 4, 5\}$. Work from the inside of the parentheses outward.

$A \cap C$

Elements in both A and C. Only 3 qualifies: $\quad A \cap C = \{3\}$

$(A \cup B) \cap C$

First the inside: $A \cup B = \{1, 2, 3, 4\}$.

Then intersect with $C = \{3, 4, 5\}$: $\quad (A \cup B) \cap C = \{3, 4\}$

$A \setminus (B \cap C)$

First the inside: $B \cap C = \{3, 4\}$.

Then remove those from $A = \{1, 2, 3\}$ — the 4 was not there to remove: $A \setminus (B \cap C) = \{1, 2\}$

Properties of Union and Intersection

These behave much like $+$ and $\times$ in ordinary algebra.

Commutative — order does not matter

$$A \cup B = B \cup A \qquad A \cap B = B \cap A$$

Associative — grouping does not matter

$$A \cup (B \cup C) = (A \cup B) \cup C \qquad A \cap (B \cap C) = (A \cap B) \cap C$$

$\cup$ and $\cap$ are **binary** — each joins two sets at a time. Associativity is what lets us drop the parentheses and write $A \cup B \cup C$, exactly as for $1 + 2 + 3$.

Distributive — each distributes over the other

$$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$$
$$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$$

The first of these has no arithmetic analogue — $+$ does not distribute over $\times$.

De Morgan's Laws for Sets

The laws from Unit 1-1 carry over word for word, with $\cup$ for OR and $\cap$ for AND.

De Morgan's Laws (sets)

$$\overline{A \cup B} = \bar{A} \cap \bar{B}$$
$$\overline{A \cap B} = \bar{A} \cup \bar{B}$$

Why they are the same laws

$$\begin{aligned} x \in \overline{A \cap B} \quad &\Leftrightarrow \quad \neg\big(x \in A \ \wedge \ x \in B\big) \\ &\Leftrightarrow \quad \neg(x \in A) \ \vee \ \neg(x \in B) \qquad \text{(logical De Morgan)} \\ &\Leftrightarrow \quad x \in \bar{A} \cup \bar{B} \end{aligned}$$

Set theory is logic with the statements collected into piles.

Back to the Cohort

Unit 1-1 opened with a criterion we could write in logic but not yet *collect*:

The English

"Patients who are diabetic AND (hypertensive OR obese), but NOT pregnant."

Let U be all patients in the study, and name each property as a statement about x:

$$d(x) = \text{“}x \text{ is diabetic”} \qquad h(x) = \text{“}x \text{ is hypertensive”}$$
$$o(x) = \text{“}x \text{ is obese”} \qquad g(x) = \text{“}x \text{ is pregnant”}$$

The cohort is a set

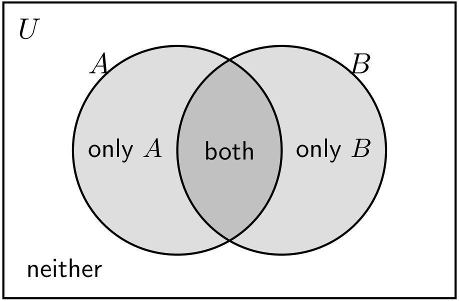
$$\text{Cohort} \;=\; \left\{\, x \in U \;\middle|\; d(x) \wedge (h(x) \vee o(x)) \wedge \neg g(x) \,\right\}$$

Logic supplies the rule; set-builder notation turns it into a **collection you can count**.

Venn Diagrams and Counting

Reading a Two-Set Venn Diagram

Two overlapping circles cut the universe into **four** regions — and every patient sits in exactly one of them.



only A	$A \setminus B$
both	$A \cap B$
only B	$B \setminus A$
neither	$\overline{A \cup B} = \bar{A} \cap \bar{B}$

Inclusion–Exclusion

How many elements are in $A \cup B$? Adding $|A|$ and $|B|$ counts the overlap **twice**, so we subtract it back off once:

Inclusion–exclusion (two sets)

$$|A \cup B| = |A| + |B| - |A \cap B|$$

Example

In a clinic of 100 patients, let D be the set of diabetics and H the set of hypertensives. Suppose $|D| = 40$, $|H| = 55$, and $|D \cap H| = 25$.

$$|D \cup H| = 40 + 55 - 25 = 70$$

So 70 patients have at least one condition — and $100 - 70 = 30$ have neither.

Notice that $40 + 55 = 95 \neq 70$. Forgetting to subtract the overlap is the classic error.

Answering the Opening Question

The question

"How many patients have diabetes **or** *hypertension, but* **not both***?"*

In symbols, that is the region $(D \cup H) \setminus (D \cap H)$ — the two crescents, but not the middle:

$$|(D \cup H) \setminus (D \cap H)| \;=\; |D \cup H| - |D \cap H|$$

With the numbers from the previous slide:

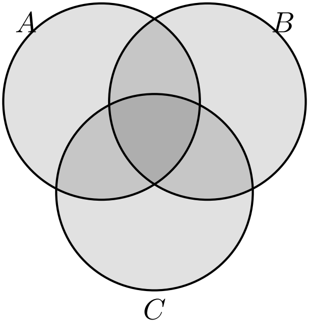
$$70 - 25 = 45$$

Recognize the shape

"Or, but not both" is the **exclusive OR** from Unit 1-1. The same idea, now counted.

Three-Set Venn Diagrams

Three circles cut the universe into **eight** regions (2^3 — one for each in/out combination, exactly like a truth table).



Strategy: fill in from the middle outward

Start with $|A \cap B \cap C|$, then the two-way overlaps (subtracting the center), then the single regions. Each region gets counted exactly once.

Exercise

Try it

A clinic surveys 120 patients. Let E be the set who exercise regularly, and D the set who follow a special diet:

$$|E| = 70, \qquad |D| = 45, \qquad |E \cap D| = 25$$

1. How many do **at least one** of the two?
2. How many do **neither**?
3. How many exercise but do **not** diet?

Exercise: Solution

1. At least one — inclusion–exclusion

$$|E \cup D| = |E| + |D| - |E \cap D| = 70 + 45 - 25 = 90$$

2. Neither — everything outside that union

$$|\overline{E \cup D}| = 120 - 90 = 30$$

3. Exercise but not diet — a difference

$$|E \setminus D| = |E| - |E \cap D| = 70 - 25 = 45$$

The 25 who do both are already counted inside E, so subtract them off.

Check the four regions: $45 + 25 + 20 + 30 = 120$ — exercise only, both, diet only, neither.

Sets of Numbers

The Standard Number Systems

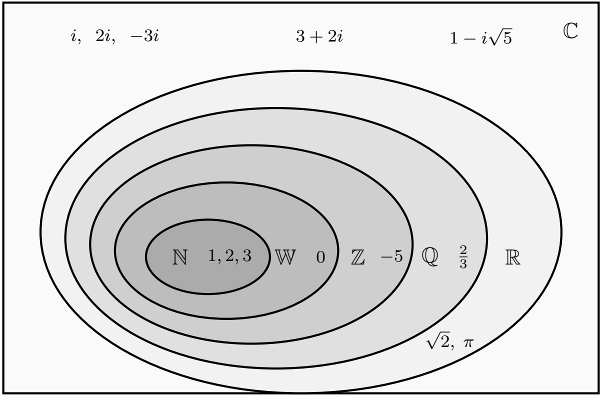
- **Natural numbers:** $\mathbb{N} = \{1, 2, 3, 4, \ldots\}$ — the counting numbers.
- **Whole numbers:** $\mathbb{W} = \{0, 1, 2, 3, 4, \ldots\}$ — the naturals, plus zero.
- **Integers:** $\mathbb{Z} = \{\ldots, -3, -2, -1, 0, 1, 2, 3, \ldots\}$ — the wholes, plus negatives.
- **Rational numbers:** $\mathbb{Q} = \left\{ \frac{p}{q} \;\middle|\; p, q \in \mathbb{Z},\ q \neq 0 \right\}$ — ratios of integers.
- **Real numbers:** $\mathbb{R}$ — every rational **and** every irrational.
- **Complex numbers:** $\mathbb{C} = \{\, a + bi \mid a, b \in \mathbb{R} \,\}$ — a real part plus an imaginary part, where $i = \sqrt{-1}$.

Each one sits inside the next

$$\mathbb{N} \subset \mathbb{W} \subset \mathbb{Z} \subset \mathbb{Q} \subset \mathbb{R} \subset \mathbb{C}$$

The imaginary numbers are **not** a link in that chain: no real number squares to a negative, so $2i \notin \mathbb{R}$. They sit inside $\mathbb{C}$ alongside $\mathbb{R}$, not around it. ($\mathbb{Z}$ is the German *Zahlen*, "numbers"; $\mathbb{Q}$ is for *quotient*.)

The Number Systems, Nested



Inside $\mathbb{R}$ but outside $\mathbb{Q}$ are the **irrationals** — $\sqrt{2}$, π — which are not ratios of integers. Inside $\mathbb{C}$ but outside $\mathbb{R}$ is everything with a nonzero **imaginary** part: i is nowhere on the real line.

Rational vs. Irrational

A rational number's decimal expansion either stops or repeats

- $\frac{1}{2} = 0.5$ (terminates)
- $\frac{1}{3} = 0.3333\ldots$ (repeats)
- $\frac{1}{7} = 0.142857\,142857\ldots$ (repeats, with period 6)

An irrational number's decimal expansion does neither

- $\sqrt{2} = 1.41421356\ldots$
- $\pi = 3.14159265\ldots$
- $e = 2.71828182\ldots$

A warning that Unit 1-4 will pay off

A computer stores only finitely many digits, so it never holds π, or even $\frac{1}{3}$, exactly. That is the subject of **floating-point arithmetic**.

Exercise

Try it

For each number below, name **every** one of $\mathbb{N}, \mathbb{W}, \mathbb{Z}, \mathbb{Q}, \mathbb{R}$ that it belongs to — put a check in each box that applies.

	$\mathbb{N}$	$\mathbb{W}$	$\mathbb{Z}$	$\mathbb{Q}$	$\mathbb{R}$
7					
-3					
0					
$\frac{5}{8}$					
$\sqrt{9}$					
$\sqrt{5}$					

Exercise: Solution

	$\mathbb{N}$	$\mathbb{W}$	$\mathbb{Z}$	$\mathbb{Q}$	$\mathbb{R}$	
7	✓	✓	✓	✓	✓	
-3			✓	✓	✓	negative
0		✓	✓	✓	✓	$\mathbb{N}$ starts at 1
$\frac{5}{8}$				✓	✓	a ratio of integers
$\sqrt{9}$	✓	✓	✓	✓	✓	$\sqrt{9} = 3$
$\sqrt{5}$					✓	irrational

The two traps

$\sqrt{9}$ is simply 3 — a square root is not automatically irrational. And 0 lands in $\mathbb{W}$ but not $\mathbb{N}$, since we take $\mathbb{N} = \{1, 2, 3, \ldots\}$.

Exercise: Implications Between Number Systems

Try it — true or false? If false, give a counterexample.

1. $x \in \mathbb{N} \;\rightarrow\; x \in \mathbb{Z}$
2. $x \in \mathbb{R} \;\rightarrow\; x \in \mathbb{Z}$
3. $x \in \mathbb{Z} \;\rightarrow\; x \in \mathbb{Q}$
4. $x \in \mathbb{Q} \;\rightarrow\; x \in \mathbb{R}$
5. $x \notin \mathbb{Q} \;\rightarrow\; x \notin \mathbb{R}$
6. $x \notin \mathbb{Z} \;\rightarrow\; x \notin \mathbb{N}$

Recall from Unit 1-1 that an implication fails only when the left side is **true** and the right side is **false** — so one counterexample settles it.

Exercise: Solution

Claim	T/F	Why
1. $x \in \mathbb{N} \to x \in \mathbb{Z}$	T	every natural number is an integer
2. $x \in \mathbb{R} \to x \in \mathbb{Z}$	F	$x = 0.5$ is real but not an integer
3. $x \in \mathbb{Z} \to x \in \mathbb{Q}$	T	$n = \frac{n}{1}$, a ratio of integers
4. $x \in \mathbb{Q} \to x \in \mathbb{R}$	T	every rational sits on the real line
5. $x \notin \mathbb{Q} \to x \notin \mathbb{R}$	F	$x = \sqrt{2}$ is irrational **and** real
6. $x \notin \mathbb{Z} \to x \notin \mathbb{N}$	T	the contrapositive of $\#1$

Notice what these really ask. "$x \in A \to x \in B$" is true for **every** x exactly when $A \subseteq B$ — so the true rows are precisely the containments in $\mathbb{N} \subset \mathbb{W} \subset \mathbb{Z} \subset \mathbb{Q} \subset \mathbb{R}$, read left to right. Reading one *right to left* ($\#2$) or negating without flipping the order ($\#5$) breaks it.

How Big Is an Infinite Set?

We defined the **cardinality** $|A|$ as the number of elements in A, so $|\{1, 2, 3\}| = 3$. What, then, is the cardinality $|\mathbb{N}|$?

Claim: no natural number is the cardinality of $\mathbb{N}$

Suppose one were — say $|\mathbb{N}| = N$ for some $N \in \mathbb{N}$.

Then $\mathbb{N} = \{1, 2, 3, \ldots\}$ would stop after N elements, so its **largest** element would be N itself.

But $N \in \mathbb{N}$ forces $N + 1 \in \mathbb{N}$ as well, and $N + 1 > N$. So N is **not** the largest element — which contradicts what we just said.

Nothing about N was special: *whatever* number we pick, adding 1 produces a bigger one still in $\mathbb{N}$. So the assumption fails for every candidate, and $|\mathbb{N}|$ is no natural number at all. ∎

Writing $|\mathbb{N}| = \infty$ does not rescue us either — we are about to find that **not all infinities are the same size**. We need a notion of "same cardinality" that never requires finishing a count, yet agrees with ordinary counting on finite sets.

Same Size Means Perfect Pairing

A theater has as many seats as ticket-holders when **everyone sits, every seat is filled, and nobody doubles up** — nothing was counted.

Definition by pairing

Two sets have the **same cardinality** if their elements can be matched one-to-one, with nothing left over on either side.

Same cardinality

$A = \{2, 4, 6\}, \quad B = \{a, b, c\}$

$$\begin{array}{ccc} A & & B \\ 2 & \longleftrightarrow & a \\ 4 & \longleftrightarrow & b \\ 6 & \longleftrightarrow & c \end{array}$$

Nothing left over: $|A| = |B| = 3$.

Different cardinality

$C = \{1, 2, 3, 4\}, \quad D = \{x, y, z\}$

$$\begin{array}{ccc} C & & D \\ 1 & \longleftrightarrow & x \\ 2 & \longleftrightarrow & y \\ 3 & \longleftrightarrow & z \\ \boxed{4} & \text{left over} & \end{array}$$

D runs out first — as does every other pairing: $|C| \neq |D|$.

The Unsettling Example

The pairing test never asks us to finish counting — so it still works when both sets are **infinite**. Pair each natural number with its double:

$$\mathbb{N} = \{1, 2, 3, 4, \ldots\} \qquad\qquad E = \{2, 4, 6, 8, \ldots\}$$

$$\begin{array}{ccc} 1 & \longleftrightarrow & 2 \\ 2 & \longleftrightarrow & 4 \\ 3 & \longleftrightarrow & 6 \\ 4 & \longleftrightarrow & 8 \\ \vdots & & \vdots \\ n & \longleftrightarrow & 2n \end{array}$$

A part can be as big as the whole

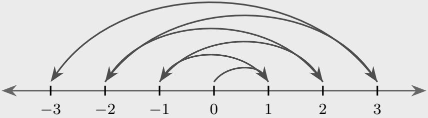
The rule $n \leftrightarrow 2n$ reaches **every** even number, and no natural number is used twice — nothing is left over on either side. So $|\mathbb{N}| = |E|$, even though E is a *proper subset* of $\mathbb{N}$: it is missing every odd number.

For a finite set that is impossible — strip out an element and the count drops. Infinite sets simply do not behave that way.

Countable Infinity: $\aleph_0$

A set is **countable** if its elements can be arranged in an infinite **list** $a_1, a_2, a_3, \ldots$ reaching every one — that is, paired off with $\mathbb{N}$. That size is $\aleph_0$, "aleph-null," the smallest infinity.

The integers: do not start at $-\infty$, zig-zag outward from 0



Reading the arcs in order gives the list $0,\ 1,\ -1,\ 2,\ -2,\ 3,\ -3,\ \ldots$

$$\begin{array}{ccc} \mathbb{N} & & \mathbb{Z} \\ 1 & \longleftrightarrow & 0 \\ 2 & \longleftrightarrow & 1 \\ 3 & \longleftrightarrow & -1 \\ 4 & \longleftrightarrow & 2 \\ \vdots & & \vdots \end{array}$$

Position n in the list pairs with the nth integer out: every integer is reached — -2 is 5th, -17 is 34th — and none twice. So $|\mathbb{Z}| = \aleph_0$.

The Rationals Are Countable Too

Every positive fraction $\frac{p}{q}$ sits in a **grid** — numerator p picks the row, denominator q the column — and sweeping the **diagonals** threads one path through all of them.

$$q \longrightarrow$$

$$\begin{array}{c|ccccc}
 & & & & & \\
 & \frac{1}{1} & \frac{1}{2} & \frac{1}{3} & \frac{1}{4} & \cdots \\
 & \frac{2}{1} & \cancel{\frac{2}{2}} & \frac{2}{3} & \frac{2}{4} & \cdots \\
p \downarrow & \frac{3}{1} & \frac{3}{2} & \frac{3}{3} & \frac{3}{4} & \cdots \\
 & \frac{4}{1} & \frac{4}{2} & \frac{4}{3} & \frac{4}{4} & \cdots \\
 & \vdots & \vdots & \vdots & \vdots & \ddots
\end{array}$$

$$\begin{array}{ccc}
\mathbb{N} & & \mathbb{Q}^+ \\
1 & \longleftrightarrow & \frac{1}{1} \\
2 & \longleftrightarrow & \frac{1}{2} \\
3 & \longleftrightarrow & \frac{2}{1} \\
4 & \longleftrightarrow & \frac{3}{1} \\
5 & \longleftrightarrow & \frac{1}{3} \\
\vdots & & \vdots
\end{array}$$

$\frac{2}{2}$ is struck out — it *is* $\frac{1}{1}$. Skipping repeats, the path reaches every positive rational once; interleaving the negatives and 0 as for $\mathbb{Z}$ gives $|\mathbb{Q}| = \aleph_0$.

Cantor's Diagonal Argument

Suppose every real between 0 and 1 could be listed too, in *some* order:

$$\begin{aligned}
r_1 &= 0.\underline{2}5819340671\ldots\\
r_2 &= 0.4\underline{9}710283561\ldots\\
r_3 &= 0.60\underline{4}82137952\ldots\\
r_4 &= 0.317\underline{6}05942813\ldots\\
r_5 &= 0.8524\underline{0}3718694\ldots\\
r_6 &= 0.19365\underline{7}240817\ldots\\
r_7 &= 0.472018\underline{3}56921\ldots\\
r_8 &= 0.5083761\underline{9}2470\ldots\\
r_9 &= 0.26194837\underline{5}186\ldots\\
r_{10} &= 0.739162840\underline{2}65\ldots\\
&\vdots
\end{aligned}$$

Build a number the list misses

Walk down the underlined diagonal and replace each digit by **any different** digit — $\underline{2} \to 7$, $\underline{9} \to 3$, $\underline{4} \to 9$, and so on:

$$d \;=\; 0.7391628405\ldots$$

The rows may sit in **any order**: nothing here used one. Shuffle them, re-read the new diagonal, and the recipe still works.

Could d be a line further down — r_{10}, say? Stack them and the first **nine** digits agree:

$$\begin{aligned}
d \quad &= 0.739162840\underline{5}\ldots\\
r_{10} &= 0.739162840\underline{2}\ldots
\end{aligned}$$

The tenth splits them, since that is the digit the rule changed. Every r_n is defeated the same way at its own nth digit — so d is on **no** line of the list.

What the Diagonal Proves

The contradiction

We assumed the list contained **every** real between 0 and 1. The diagonal hands us one it missed. So no such list exists — and it is not a matter of trying harder, since the recipe defeats *any* list you propose.

The reals are uncountable

$\mathbb{R}$ cannot be paired off with $\mathbb{N}$. Its cardinality is strictly larger, and is written $\mathfrak{c}$ — "the continuum":

$$\aleph_0 \;=\; |\mathbb{N}| = |\mathbb{Z}| = |\mathbb{Q}| \;<\; |\mathbb{R}| \;=\; \mathfrak{c}$$

Notice where the jump happens: adding negatives ($\mathbb{Z}$) and all the fractions ($\mathbb{Q}$) costs nothing. Adding the **irrationals** changes the size of infinity.

The Gap Between the Two

An honest open question

Is there a set strictly bigger than $\mathbb{N}$ but strictly smaller than $\mathbb{R}$ — some size wedged between $\aleph_0$ and $\mathfrak{c}$?

The guess that there is **no** such size is the **continuum hypothesis**.

The answer is stranger than yes or no

Gödel and Cohen proved that the continuum hypothesis can be neither **proved** nor **disproved** from the standard axioms of set theory. Both answers are consistent. You may assume either one.

A reminder that "how many?" is a deeper question than it looks — and that the number line you have used since grade school still holds open problems.

Intervals

Intervals, and How We Write Them

Interval

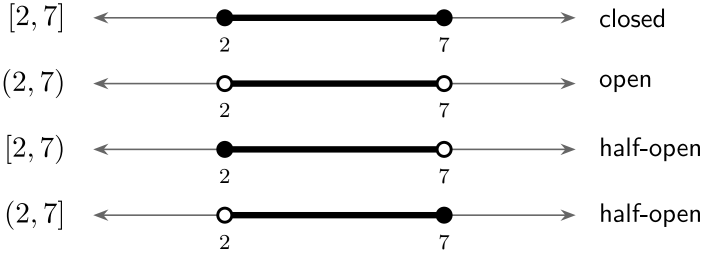
An **interval** is the set of **all** real numbers lying between two given numbers a and b, called the **endpoints**. The bracket records whether each endpoint is itself included:

$$\begin{aligned} [a,b] &= \{\, x \in \mathbb{R} \mid a \le x \le b \,\} \\ (a,b) &= \{\, x \in \mathbb{R} \mid a < x < b \,\} \\ [a,b) &= \{\, x \in \mathbb{R} \mid a \le x < b \,\} \\ (a,b] &= \{\, x \in \mathbb{R} \mid a < x \le b \,\} \end{aligned}$$

The three names

- **Closed**, $[a,b]$ — **both** endpoints included.
- **Open**, (a,b) — **neither** endpoint included.
- **Half-open**, $[a,b)$ or $(a,b]$ — one of each (also called **half-closed**).

Each endpoint decides on its own, so two endpoints give $2 \times 2 = 4$ intervals. On 2 and 7:



Only the endpoints differ — every number strictly between 2 and 7 belongs to all four. And the notations agree endpoint by endpoint:

Bracket	**Dot**	**Inequality**	**The endpoint itself**
square, $[\,]$	filled ●	$\leq, \geq$ (bar)	**is** in the set
round, $(\,)$	open ○	$<, >$ (no bar)	is **not** in the set

Every Interval, Three Ways

Interval	**Number line**	**Set-builder**
(a, b)		$\{x \in \mathbb{R} \mid a < x < b\}$
$(a, b]$		$\{x \in \mathbb{R} \mid a < x \leq b\}$
$[a, b)$		$\{x \in \mathbb{R} \mid a \leq x < b\}$
$[a, b]$		$\{x \in \mathbb{R} \mid a \leq x \leq b\}$
(a, ∞)		$\{x \in \mathbb{R} \mid x > a\}$
$[a, \infty)$		$\{x \in \mathbb{R} \mid x \geq a\}$
$(-\infty, b)$		$\{x \in \mathbb{R} \mid x < b\}$
$(-\infty, b]$		$\{x \in \mathbb{R} \mid x \leq b\}$
$(-\infty, \infty)$		$\mathbb{R}$

∞ is **not a number** — just shorthand for "without end," so its side always takes a **round** bracket.

Exercise: Translating Between the Three

Try it — each row is filled in *one* way. Fill in the other three.

	Number line	**Interval**	**Inequality**	**Set-builder**
1.		$[-3, 5)$		
2.			$-1 < x \leq 6$	
3.				$\{x \in \mathbb{R} \mid x \geq 4\}$
4.				

Watch each endpoint separately — they are independent decisions.

Exercise: Solution

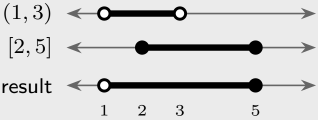
	Number line	Interval	Inequality	Set-builder
1.		$[-3, 5)$	$-3 \leq x < 5$	$\{x \in \mathbb{R} \mid -3 \leq x < 5\}$
2.		$(-1, 6]$	$-1 < x \leq 6$	$\{x \in \mathbb{R} \mid -1 < x \leq 6\}$
3.		$[4, \infty)$	$x \geq 4$	$\{x \in \mathbb{R} \mid x \geq 4\}$
4.		$(-\infty, 0)$	$x < 0$	$\{x \in \mathbb{R} \mid x < 0\}$

#3 and #4 are unbounded, so the infinite side takes a **round** bracket — there is no endpoint there to include. In #4 the open dot at 0 is what makes it $<$ rather than $\leq$.

Union of Intervals

Draw both, then read off everything covered by **either** one.

Bounded: $(1,3) \cup [2,5] = (1,5]$



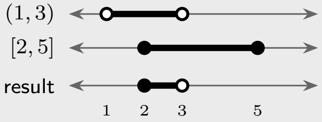
They overlap, so they merge into one stretch: open at 1, closed at 5.

Unbounded: $(-\infty,2] \cup (3,\infty)$ — a union that does *not* merge

No overlap, so nothing merges — the union is genuinely two pieces, and this is already its simplest form. An "abnormal lab value" set looks exactly like this: too low, or too high.

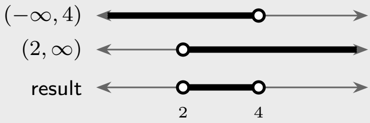
Intersection of Intervals

Bounded: $(1, 3) \cap [2, 5] = [2, 3)$



Only the overlap survives. 2 lies in **both**, so it is included; 3 is in neither, so it is not.

Unbounded: $(-\infty, 4) \cap (2, \infty) = (2, 4)$

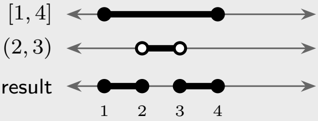


Two infinite sets can meet in a bounded one. Neither endpoint is in both, so both stay open.

Difference and Complement

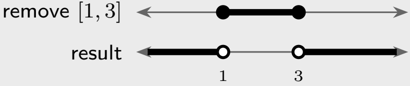
$A \setminus B$ keeps what is in A but **not** in B; the **complement** is the case $\mathbb{R} \setminus A$.

Bounded: $[1,4] \setminus (2,3) = [1,2] \cup [3,4]$



Removing an *open* middle leaves two *closed* pieces — 2 and 3 were never in $(2,3)$, so they were never removed.

Unbounded: $\mathbb{R} \setminus [1,3] = (-\infty, 1) \cup (3, \infty)$



Removing a *closed* interval leaves *open* ends: 1 and 3 were inside, so they are gone.

Watch the endpoints — every mistake happens there.

Ordered Pairs and Tuples

When Order Matters

In a set, order is irrelevant: $\{2, 5\} = \{5, 2\}$. Sometimes that is exactly wrong.

Ordered pair

An **ordered pair** (a, b) is a collection of two elements in which **the order matters**:

$$(a, b) = (c, d) \quad \Longleftrightarrow \quad a = c \text{ and } b = d$$

The one you already know: points on an xy graph

Every point you have ever plotted is an ordered pair — first entry the x-coordinate, second the y-coordinate. So $(2, 5)$ means *right 2, up 5*, while $(5, 2)$ means *right 5, up 2*: two different places from the same two numbers.

Notation warning

$(2, 5)$ is also how we write an open interval. **Context tells you which is meant** — a pair of coordinates, or a stretch of the number line.

Tuples

n-tuple

An n-**tuple** $(a_1, a_2, \ldots, a_n)$ generalizes the ordered pair to any finite number of entries, order still mattering.

- A **2-tuple** is an ordered pair: (a, b).
- A **3-tuple** (triple): (a, b, c).
- A **4-tuple**: (a, b, c, d).   And so on.

Two n-tuples are equal exactly when they agree entry by entry: $a_i = b_i$ for every i.

Examples

- $(2, 5) \neq (5, 2)$ — order matters.
- $(1, 2, 3) \neq (3, 2, 1)$ — same reason.
- $(x, y, z) = (4, -2, 7)$ locates a point in three-dimensional space.

Why Tuples Matter Here

They are the bridge to the rest of the course

- **Coordinates:** every point on a graph is a tuple.
- **Relations and functions (Unit 1-3):** a relation is literally a *set of ordered pairs* $(input, output)$.
- **Vectors (Unit 2):** a vector in $\mathbb{R}^n$ is an n-tuple of numbers.
- **Data:** one row of a data table — one patient's measurements — is a tuple. A whole dataset is a set of tuples.

The takeaway

A **set** answers "which things?" A **tuple** answers "which things, in which slots?" Nearly all of data science is built out of the two.

Review

Review

What we covered

1. A **set** is a collection of elements; $\{x \mid \ldots\}$ describes one by a rule.
2. $A \subseteq B$ means every element of A is in B; $\emptyset$ is a subset of everything.
3. $A \cup B$, $A \cap B$, $A \setminus B$, $\bar{A}$ are the set versions of OR, AND, AND NOT, NOT — **De Morgan** included.
4. **Venn diagrams** organize the regions; $|A \cup B| = |A| + |B| - |A \cap B|$ counts them.
5. $\mathbb{N} \subset \mathbb{W} \subset \mathbb{Z} \subset \mathbb{Q} \subset \mathbb{R}$; rational decimals terminate or repeat, irrational ones do not.
6. $\mathbb{N}, \mathbb{Z}, \mathbb{Q}$ are all **countable** ($\aleph_0$); Cantor's diagonal shows $\mathbb{R}$ is strictly bigger.
7. **Intervals** are stretches of the real line; brackets record the endpoints.
8. **Tuples** are collections where order matters — points, data rows, vectors.

Notebooks: `U1-2_Sets-1_Operations`, `-2_VennDiagrams`, `-3_NumberSystems`, `-4_Intervals`